

DATA EXPLORATION REPORT

Retail Sales & Customer Retention Analysis Project.

1. Dataset Overview

The dataset used for this project is a 6-month transactional record from Local Supermarket covering the period December 2025 through May 2026. It captures every line item sold across all product categories, customer visits, and payment channels. The dataset is clean, complete, and structurally ready for analysis.

63,571 Total Line Items <i>Rows in dataset</i>	13,317 Unique Bills <i>Transaction_Id</i>	120 Customers <i>Unique Customer_Id</i>	35 Products <i>Across 10 categories</i>
₹1.05 Cr Total Net Revenue <i>6-month period</i>	₹795 Avg Transaction Value <i>Per bill</i>	8.9 items Avg Basket Size <i>Per transaction</i>	17 Missing Values <i>99% complete</i>

2. Sales Data Exploration

2.1 Monthly Revenue Trend

Revenue is tracked across all 6 months. The following distribution was observed from the dataset:

Month	Net Revenue (₹)	vs. Period Average	Observation
December 2025	17,65,148	0	Solid opening month; pre-Pongal stock-up begins
January 2026	18,74,131	+2.9%	Highest month — Pongal festival (14–15 Jan) spike
February 2026	15,90,456	−12.8%	Shortest month; expected dip
March 2026	17,90,511	−1.7%	Steady recovery; month-end buying evident
April 2026	17,33,484	−4.8%	Tamil New Year (14 Apr) provides mid-month boost
May 2026	18,37,185	+0.9%	Strong close; near-peak performance

- Peak month:** January 2026 — ₹18.74 Lakh driven by Pongal festival purchases
- Low month:** February 2026 — ₹15.90 Lakh (calendar effect — only 28 days)
- Revenue range:** ₹15.9L – ₹18.7L across the period — low volatility, stable business
- 6-month total:** ₹1,05,90,915 — approximately ₹1.06 Crore

2.2 Category Performance

10 product categories contribute to total revenue. The breakdown by transaction volume is as follows:

Category	Line Items	Share (%)	Exploration Priority
Staples	9,004	14.16%	Highest volume — core of business
Spices	8,996	14.15%	Very close second — high repeat purchase
Oils & Fats	7,442	11.71%	High unit value — significant revenue driver
Beverages	7,398	11.64%	Consistent daily purchase category
Snacks	7,204	11.33%	High frequency, lower unit value
Dairy	7,119	11.20%	Loyalty-sensitive — perishable repurchase
Vegetables	5,470	8.6%	Price-volatile — good for trend analysis
Personal Care	5,442	8.56%	Strong brand loyalty signals
Household	3,706	5.83%	Lower frequency — bulk purchase patterns
Fruits	1,790	2.82%	Lowest — seasonal and supplementary

2.3 Top 10 Products by Revenue

Rank	Product Name	Category	Revenue (₹)	Insight
1	Ponni Boiled Rice 5kg	Staples	9,25,139	Staple anchor product — every household
2	Atta Wheat Flour 5kg	Staples	7,99,082	Second-highest — consistent daily demand
3	Pure Ghee 200ml	Oils & Fats	7,27,808	High unit price drives revenue despite lower qty
4	Groundnut Oil 1L	Oils & Fats	6,06,108	Tamil Nadu household staple oil
5	Coconut Oil 500ml	Oils & Fats	5,71,502	Dual-use (cooking + personal care)
6	Horlicks 200g	Beverages	5,45,632	Premium health drink — loyalty purchase
7	Detergent Powder 1kg	Household	4,95,893	High unit price compensates lower volume
8	Toor Dal 1kg	Staples	4,85,301	Essential dal — daily cooking staple
9	Sunflower Oil 1L	Oils & Fats	4,63,691	Value alternative to groundnut oil
10	Coffee Powder 200g	Beverages	4,28,749	Strong South India brand preference

2.4 Transaction & Basket Analysis

Metric	Value	Interpretation
Average Transaction Value (ATV)	₹795	Healthy basket — customers buying across categories
Median Transaction Value	₹684	Skewed right — some high-value bulk purchases
Maximum Transaction Value	₹4,201	Large monthly stock-up transactions
Minimum Transaction Value	₹19	Single small item purchases (e.g. spice sachet)
Average Basket Size	8.9 items	Multi-category shopping per visit
Median Basket Size	8 items	Consistent multi-item buying behavior
Maximum Basket Size	38 items	Bulk/festive occasion purchases

2.5 Payment Method Distribution

Payment Method	Line Items	Share (%)	Exploration Note
UPI	28,441	44.7%	Dominant — reflects digital payment adoption
Cash	28,185	44.3%	Nearly equal to UPI
Card	6,962	11.0%	Minority share — larger purchases tend to use card

2.6 Cashier & Operations

Cashier ID	Line Items Processed	Share (%)	Note
EMP01	23,813	37.46%	Marginally highest — possible peak shift assignment
EMP02	13,238	20.82%	Balanced load across all four staff members
EMP03	13,356	21.01%	Equitable distribution — good scheduling
EMP04	13,164	20.71%	Slightly lower — may cover fewer weekend shifts

Conclusion & Credits:

This report highlights key insights from retail sales data using SQL, focusing on sales trends, customer behavior, and operational performance. Created as a hands-on project to demonstrate practical data analysis skills using PostgreSQL.